Rejection Sampling for Medical MCQ Generation and Fine-Tuning TinyLlaMA for Layperson QA

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PowerPoint Slides: PowerPoint Slides
Video Presentation: Video Presentation
Code Repository: github.com/chelseanbr/aih-project
Data Source: huggingface.co/datasets/lavita/MedQuAD

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1 Introduction

Medical question answering is a challenging task due to the complexity of medical language and the need for accurate information. In this project, I generate multiple-choice questions (MCQs) from the MedQuAD layperson question answering dataset using rejection sampling. The objective is to create a set of MCQs that can effectively assess layperson medical knowledge. Additionally, I perform supervised fine-tuning of the TinyLLaMA model using the curated MCQs to improve its performance in answering layperson medical questions.

2 Related Work

Medical question answering has been explored as a structured retrieval and reasoning task in prior work. Ben Abacha and Demner-Fushman [1] introduced an entailment-based framework for medical question answering, highlighting the importance of question relevance and answer justification when designing systems for layperson use. Recent advances in scaling language models have also improved the ability to reason over complex domains like mathematics and medicine. Yuan et al. [2] studied how scaling model size impacts performance on mathematical reasoning tasks, showing that larger models exhibit stronger generalization with limited additional supervision. In the context of data generation for medical QA, Zhu et al. [3] proposed CURA, a method for generating synthetic QA datasets using large language models. Their work shows that careful control of synthetic data generation can improve downstream QA performance, particularly in specialized domains like healthcare. Building on these directions, this project focuses on generating high-quality MCQs from existing layperson medical QA datasets and fine-tuning smaller language models for efficient and accurate question answering.

3 Methodology

The methodology consists of two main components: MCQ generation and model fine-tuning. For MCQ generation, I use a rejection sampling technique to create multiple-choice questions from the MedQuAD layperson medical dataset. This approach ensures that the generated questions are relevant, high-quality, and suitable for assessing layperson medical knowledge. For fine-tuning, I train the TinyLLaMA model on the curated MCQ dataset using supervised learning. The model learns to predict the correct answer given a medical question and multiple options. I evaluate the fine-tuned model on a separate hold-out test set to assess its performance on layperson medical question answering. An overview of the MCQ generation and fine-tuning workflow is illustrated in Figure 1.

4 Results

The results evaluate the performance of the fine-tuned TinyLLaMA model against the baseline TinyLLaMA and MedAlpaca models on the layperson medical MCQ dataset.

The overall workflow for MCQ generation and model fine-tuning is illustrated in Figure 1.

4.1 Model Answer Distributions

The answer distributions for each model are shown in Figures 3, 4, and 5. The baseline TinyLLaMA model (Figure 3) exhibits a strong bias toward selecting a single answer choice ("B"), reflecting poor generalization to the diverse MCQ set. After supervised fine-tuning, the TinyLLaMA model (Figure 4) demonstrates improved alignment with the ground truth distribution, though some imbalance across options remains. The MedAlpaca 7B model (Figure 5) shows a broader spread across answer choices compared to the baseline, but still underperforms the fine-tuned TinyLLaMA on accuracy.

4.2 Model Accuracy

Figure 8 summarizes the accuracy of each model on the test set. The fine-tuned TinyLLaMA achieves the highest accuracy at 33.98These results highlight the effectiveness of using rejection-sampled MCQs and lightweight supervised fine-tuning for improving model performance on layperson medical QA tasks.

4.3 Dataset Characteristics and Training Dynamics

The distribution of answers in the training and test sets is shown in Figure 6, confirming that the dataset is balanced across answer choices A, B, C, and D. Figure 7 presents the training and validation loss curves during fine-tuning, showing a steady decrease across epochs and suggesting effective learning without signs of overfitting.

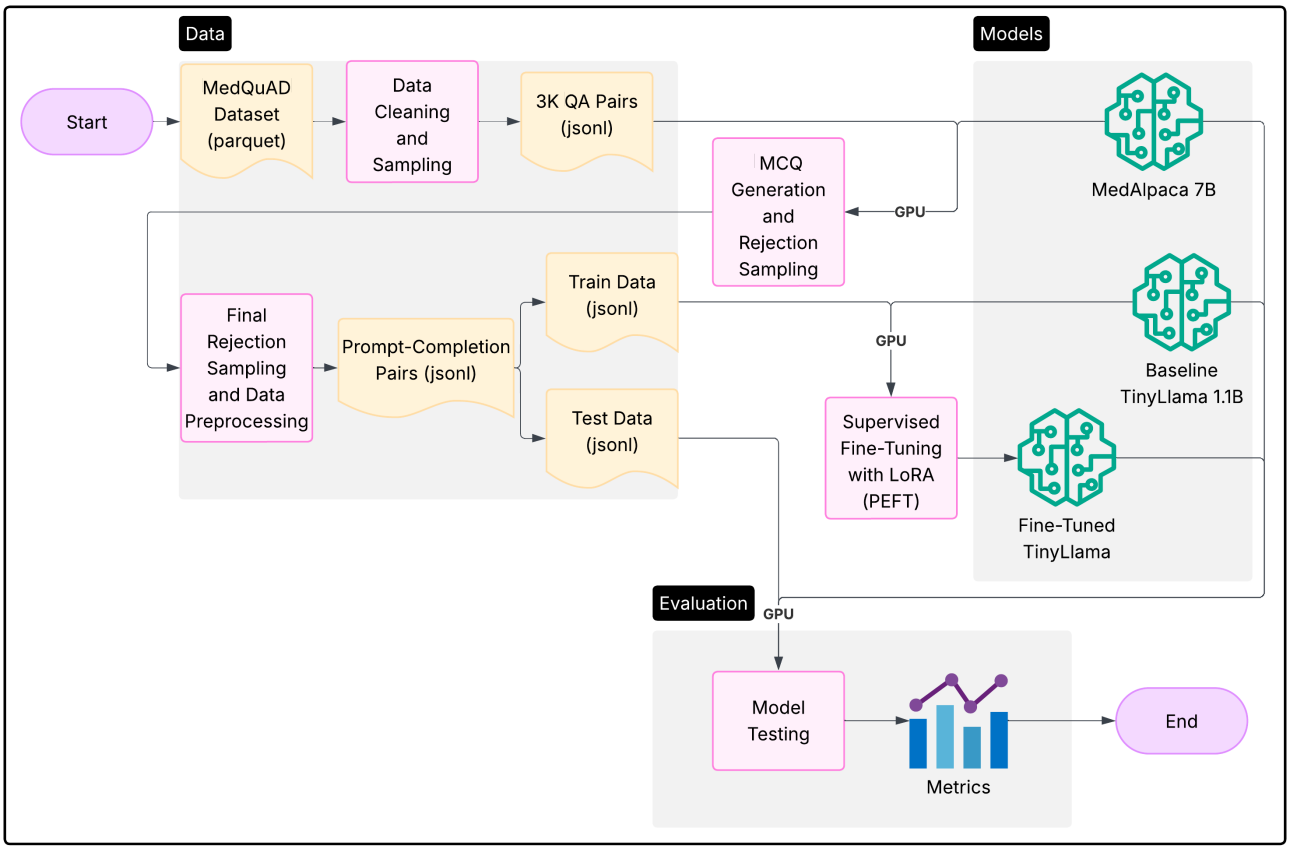

Figure 1: Workflow for MCQ Generation and Fine-Tuning

```
# Few-shot example to encourage structure
few_shot_example = """
Question: What is asthma?
Answer: Asthma is a lung condition that causes airway inflammation and breathing difficulty.
<choices>
A. A skin condition
B. A viral infection
C. A lung disease that causes airway inflammation
D. A type of heart failure
</choices>
<reason>Asthma is a lung condition that causes airway inflammation and breathing difficulty.</reason>
<answer>C</answer>
"""

# Format QA into prompt
def format_prompt(question, answer):
    return f"""
Create layperson multiple-choice answer choices and reasoning for the given medical Q&A pair.
Include 4 total choices: one correct and three plausible, but incorrect. Follow the XML-style structure below:
{few_shot_example}
Now it's your turn:
Question: {question}
Answer: {answer}
"""
```

Figure 2: MCQ Generation Prompt


```
{"question":"What causes Gum (Periodontal) Disease?","answer":"Gum disease is
caused by dental plaque -- a sticky film of bacteria that builds up on teeth.
Regular brushing and flossing help get rid of plaque. But plaque that is not
removed can harden and form tartar that brushing doesn't clean. Only a
professional cleaning by a dentist or dental hygienist can remove tartar."}
```


Figure 3: Sampled Data


```
{"question":"What causes Gum (Periodontal) Disease?","mcq_choices":"A. Genetic
mutations\nB. Poor dental hygiene\nC. Injury to the gums\nD. Periodontal
abscesses","mcq_reason":"Gum disease is caused by dental plaque -- a sticky film
of bacteria that builds up on teeth. Regular brushing and flossing help get rid
of plaque. But plaque that is not removed can harden and form tartar that
brushing doesn't clean. Only a dentist or dental hygienist can remove
tartar.","mcq_answer":"B"}
```


Figure 4: Generated Data


```
{"prompt":"You are a helpful and concise medical assistant.\n\nAnswer the
following multiple-choice question. First, briefly explain your reasoning using
relevant medical knowledge. Then state your final answer letter using exactly one
<answer></answer> tag.\n\nInstructions:\n- Use only one <answer> tag (e.g.,
<answer>B</answer>)\n- Choose the best answer letter\n- Stop immediately after
the </answer> tag\n\n<question>\nWhat causes Gum (Periodontal) Disease?\nA.
Genetic mutations\nB. Poor dental hygiene\nC. Injury to the gums\nD. Periodontal
abscesses\n</question>\n","completion":"Gum disease is caused by dental plaque --
a sticky film of bacteria that builds up on teeth. Regular brushing and flossing
help get rid of plaque. But plaque that is not removed can harden and form tartar
that brushing doesn't clean. Only a dentist or dental hygienist can remove
tartar.\n<answer>B</answer>"}
```


Figure 5: Final Data

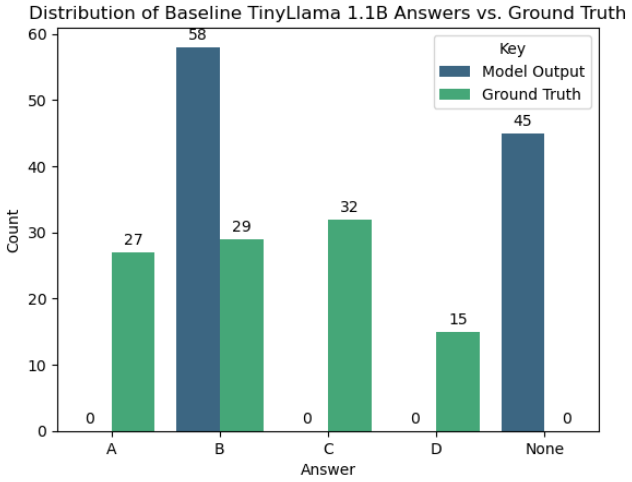

Figure 6: Baseline TinyLlama Results

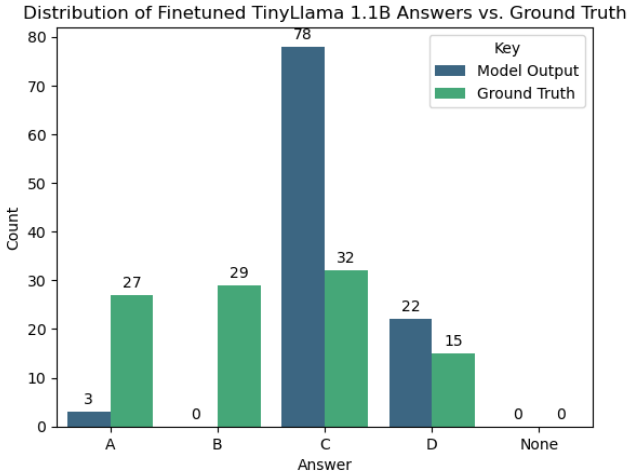

Figure 7: Finetuned TinyLlama Results

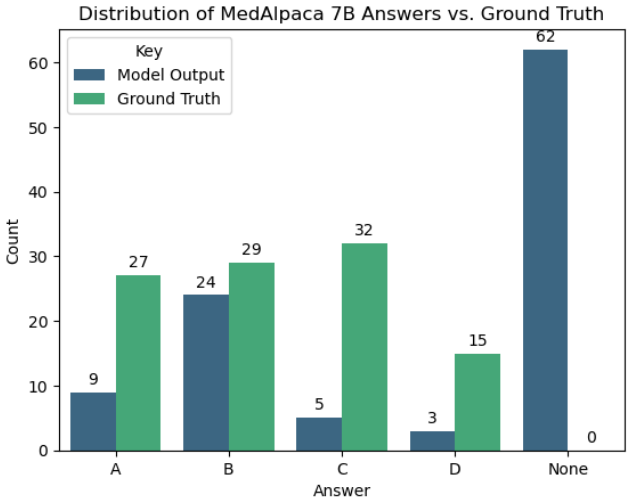

Figure 8: MedAlpaca Results

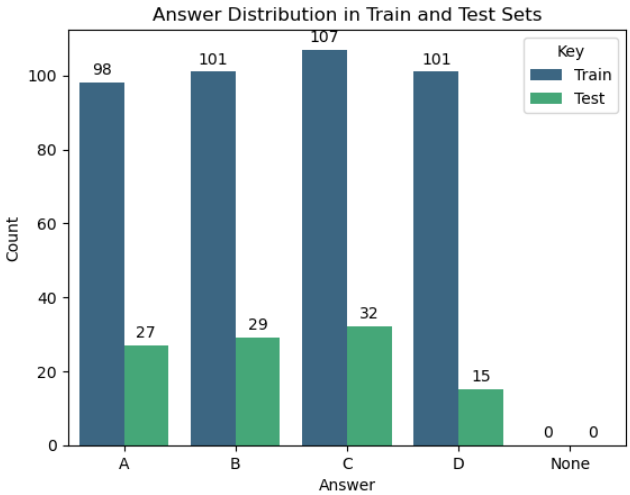

Figure 9: Train and Test Distribution

[48/48 06:11, Epoch 3/3]

Epoch	Training Loss	Validation Loss
1	0.846100	0.489880
2	0.370500	0.475589
3	0.363300	0.481885

Figure 10: Loss during Fine-Tuning

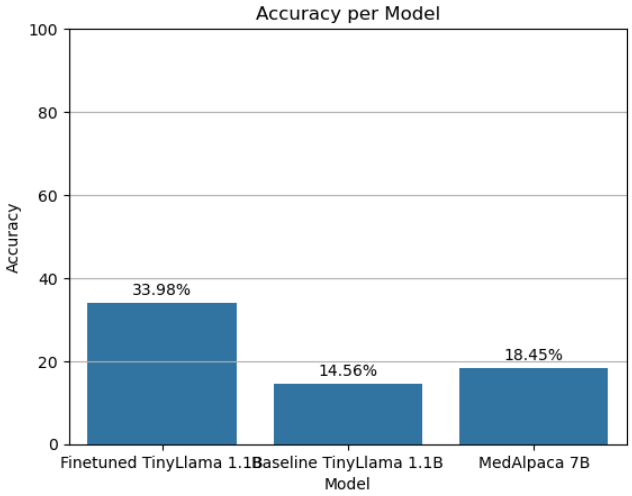

Figure 11: Accuracy per Model

5 Conclusion

In this project, I developed a pipeline for generating high-quality multiple-choice questions (MCQs) from the MedQuAD layperson medical dataset using rejection sampling, and fine-tuned the TinyLLaMA model for layperson medical question answering.

The fine-tuned TinyLLaMA model achieved a substantial improvement in accuracy over the baseline TinyLLaMA and MedAlpaca models, demonstrating the effectiveness of lightweight supervised fine-tuning combined with carefully curated training data.

These results suggest that even small language models can benefit significantly from targeted fine-tuning with high-quality, domain-specific MCQs. Future work could explore scaling this approach to larger models, incorporating additional question generation techniques, or evaluating model performance across different types of medical question formats beyond multiple-choice.

References

[1] Asma Ben Abacha and Dina Demner-Fushman. 2019. A Question-Entailment Approach to Question Answering. *BMC Bioinformatics* 20, 1 (2019). https://bmcbioinformatics.biomedcentral.com/articles/10.1186/s12859-019-3119-4 Article number 511:1–511:23.

[2] Zheng Yuan, Hongyi Yuan, Chengpeng Li, Guanting Dong, Keming Lu, Chuanqi Tan, Chang Zhou, and Jingren Zhou. 2023. Scaling Relationship on Learning Mathematical Reasoning with Large Language Models. *arXiv preprint arXiv:2308.01825* (2023). https://arxiv.org/abs/2308.01825

[3] Jiean Zhu. 2024. Cura-LLaMA: Evaluating Open-Source Large Language Model's Question Answering Capability on Medical Domain. *Applied and Computational Engineering* 90 (2024), 52–60. doi:10.54254/2755-2721/90/20241725